

AFRITEC LTD

African Information & Telecommunication Company

Building a Production-Grade Private Cloud Cluster from Bare Metal

A Full-Lifecycle IaaS Deployment Using Canonical MicroCloud on Cisco UCS Infrastructure

☐ INSERT SCREENSHOT HERE

Capture: your Mission Control Grafana dashboard with live data — the hero image for the cover.

Figure 1. Completed cluster — AFRITEC MicroCloud Mission Control dashboard (live).

Technical Project Report & Deployment Guide

Prepared by: Horus Ngu Djomou Sielatshom — Network Technician
Public Portfolio Edition · Version 1.0 · June 2026

PORTFOLIO EDITION — SANITIZED FOR PUBLIC SHARING

Table of Contents

To update this table in Microsoft Word: right-click anywhere in it and choose "Update Field" → "Update entire table."

1. Executive Summary

Note This is a sanitized portfolio edition. Internal IP addresses, hostnames, and organization-specific details have been replaced with generic examples for public sharing. The technical methodology, architecture, and troubleshooting content are complete and accurate.

This report documents the complete design, deployment, and operation of a production-grade private cloud platform built for AFRITEC Ltd. The platform transforms three bare-metal Cisco UCS C220 M5 rack servers into a fully functional, highly available Infrastructure-as-a-Service (IaaS) environment using Canonical's MicroCloud technology stack.

The solution delivers three core cloud capabilities — compute, distributed storage, and software-defined networking — in a single integrated cluster, complemented by a real-time monitoring and observability layer. The result is an on-premises cloud comparable in function to public-cloud offerings, operated entirely on infrastructure owned and controlled by AFRITEC, using open-source software.

1.1 Business Outcomes

- **Self-hosted IaaS platform:** capable of running virtual machines and containers on demand, eliminating dependency on external cloud providers for internal workloads.
- **High availability:** workloads and data are distributed across three nodes; the failure of any single node does not cause data loss or total outage.
- **Distributed storage:** approximately 19 TiB of replicated block storage provided by a Ceph cluster spanning 17 storage devices.
- **Full observability:** every node is monitored in real time through a custom operations dashboard, enabling proactive management.
- **Secure remote operation:** the platform can be managed securely from outside the office network via an encrypted mesh VPN.

1.2 Solution at a Glance

Capability	Technology	Business Value
Compute	LXD	Run VMs & containers on demand
Distributed Storage	MicroCeph (Ceph)	Resilient, replicated storage pool
Networking	MicroOVN (OVN)	Isolated, software-defined tenant networks
Orchestration	MicroCloud	Single-command cluster management
Monitoring	Prometheus + Grafana	Real-time health & capacity visibility
Remote Access	Tailscale (WireGuard)	Secure management from anywhere

2. Project Overview

2.1 Objective

Deploy a three-node private cloud cluster using Canonical MicroCloud on Cisco UCS C220 M5 servers, providing AFRITEC Ltd with a self-hosted IaaS platform for compute, storage, and networking, fully monitored and securely accessible for remote operation.

2.2 What is MicroCloud?

MicroCloud is an open-source project by Canonical that packages four components into a single automated deployment: **LXD** (a manager for containers and virtual machines), **MicroCeph** (distributed block storage based on Ceph), **MicroOVN** (software-defined networking based on Open Virtual Network), and the **MicroCloud orchestrator** that ties them together.

Conceptually, MicroCloud is a lightweight alternative to OpenStack. It delivers the same fundamental IaaS building blocks — compute, storage, and networking — but with dramatically less deployment complexity. Where a traditional OpenStack deployment can require dedicated infrastructure teams and weeks of effort, MicroCloud can bring bare-metal servers to a working cloud in under an hour through a guided setup process.

2.3 Why This Approach

For a three-server footprint, MicroCloud is the appropriate technology choice. It is purpose-built for small-scale edge and on-premises deployments (3 to 50 nodes), automates the integration of its components, and provides production-grade capabilities without the operational overhead of a full OpenStack control plane. This makes it ideal for an organization seeking the benefits of private cloud without a large dedicated cloud-operations team.

2.4 Technology Stack & Versions

Component	Technology	Version	Channel
Orchestrator	MicroCloud	2.1.3	candidate
Compute	LXD	6.x	latest/stable
Storage	MicroCeph	19.2.x	candidate
Networking	MicroOVN	24.03.x	candidate
Operating System	Ubuntu	24.04 LTS	—
Metrics Collection	Prometheus	2.51.0	—
Visualization	Grafana	10.4.2	—
Metrics Agent	Node Exporter	1.7.0	—
Remote Access	Tailscale	1.9x	—

3. System Architecture

3.1 Hardware Foundation

The cluster is built on three identical Cisco UCS C220 M5SX rack-mount servers. Each server provides dual Intel Xeon processors (40 cores total), 64 GB of RAM, and ten SAS hard drives. Out-of-band management is provided by the Cisco Integrated Management Controller (CIMC) embedded in each server, accessible over a dedicated management network.

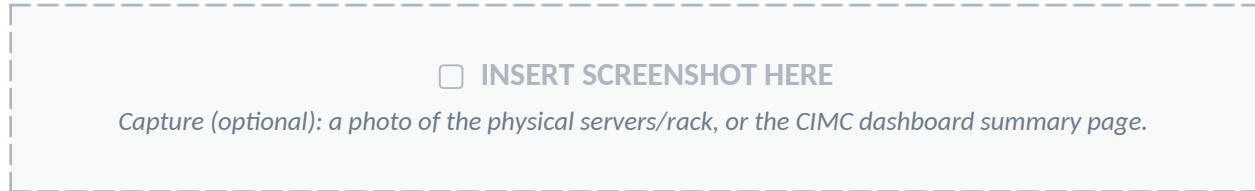


Figure 2. Cisco UCS C220 M5 servers in the AFRITEC rack.

3.2 Node Roles & Addressing

Property	SVR01	SVR02	SVR03
OS IP Address	10.0.0.11	10.0.0.12	10.0.0.13
CIMC (Mgmt) IP	10.0.1.11	10.0.1.12	10.0.1.13
Cluster Role	Database Leader	Database Voter	Database Voter
Ceph OSDs	0 (pending)	9	8
Hosted Service	Prometheus	Grafana	—
CPU / RAM	40 c / 64 GB	40 c / 64 GB	40 c / 64 GB

3.3 Storage Design

Each server separates its storage into two roles. One disk is configured as the operating-system drive. The remaining disks are presented to the Ceph storage system as individual devices, each becoming an Object Storage Daemon (OSD). Ceph then pools these OSDs across all nodes into a single unified, replicated storage cluster. This design ensures that data written by any workload is automatically distributed and protected across multiple physical servers.

3.4 Network Design

The platform uses a layered network model. A physical management network carries CIMC out-of-band traffic. A separate data network carries the servers' operating-system traffic and cluster communication. On top of this, MicroOVN provides virtual, software-defined networks for workloads, with a defined address range reserved for instances. A dedicated bridge network with address translation provides outbound internet access for service containers.

3.5 Service Endpoints

Service	Endpoint	Purpose
LXD Web UI	https://10.0.0.11:8443	Cluster & instance management
Grafana	http://10.0.0.12:3000	Monitoring dashboards
Prometheus	http://10.0.0.11:9090	Metrics database & queries
Node Exporter (each)	http://<node-ip>:9100	Per-node system metrics

 INSERT SCREENSHOT HERE

Capture: terminal output of 'sudo microceph status' and 'sudo ceph -s' on SVR01 showing HEALTH_OK.

Figure 3. Cluster health — all three nodes online with 17 OSDs.

4. Deployment Methodology

The platform was deployed in seven sequential phases. Each phase is presented below with its objective, key steps, and the commands required to reproduce it. This section is written so that an engineer with no prior cloud experience can follow it end-to-end on comparable hardware.

Phase 1 — Hardware Preparation (CIMC & RAID)

Objective: prepare each server's storage controller so that one disk hosts the operating system and the remaining disks are available to Ceph as individual devices.

Key Steps

1. Connect to the target server's CIMC web interface using its management IP.
2. Navigate to the storage controller and create a single-disk virtual drive for the operating system; mark it as the boot drive.
3. Set every remaining disk to pass-through (JBOD) mode so Ceph can claim each one directly.
4. Verify: the OS virtual drive shows as optimal and bootable; all other disks show as JBOD, healthy.
5. Repeat for all three servers.

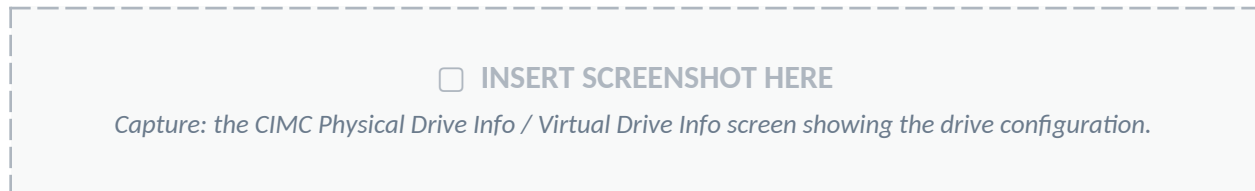


Figure 4. CIMC storage controller — OS virtual drive and JBOD disks configured.

KEY LESSON On Cisco UCS, the CIMC management plane and the server data plane are separate. CIMC can display and configure disks even when the host operating system cannot see them. Always confirm disk visibility from a live OS session, not just from CIMC.

Phase 2 — Operating System Installation

Objective: install Ubuntu 24.04 LTS on the operating-system drive of each server.

Key Steps

1. Prepare a bootable USB drive with Ubuntu 24.04 LTS (GPT partition scheme, UEFI target).
2. Boot each server from the USB drive in UEFI mode (confirm the UEFI: prefix on the boot entry).
3. During installation, select ONLY the operating-system virtual drive as the install target. Do not erase the JBOD disks reserved for Ceph.
4. Set a unique hostname per server and create the administrative user.
5. After installation, update the system:


```
sudo apt update && sudo apt upgrade -y
```

6. Repeat for all three servers.

KEY LESSON On servers presenting a mix of RAID and JBOD disks, the operating-system drive may not appear as the first device. Always identify the correct install target by size and model, never assume it is the first disk.

Phase 3 — Network Configuration

Objective: give each server a static IP address on the data network so the nodes can communicate and reach the internet.

Key Steps

1. Connect a physical network cable to the data-plane port on each server (this is separate from the CIMC management port).
2. Configure a static IP using Netplan. Example for one node:

```
network:
  version: 2
  ethernets:
    <interface>:
      dhcp4: no
      addresses: [10.0.0.11/24]
      routes:
        - to: default
          via: 10.0.0.1
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4]
```

3. Apply and verify connectivity:

```
sudo netplan apply
ping -c 3 8.8.8.8
```

4. Confirm all three nodes can ping each other before proceeding.

KEY LESSON The CIMC management IPs are not the operating-system IPs. Each server requires a separate physical network connection and IP for its OS. Confusing the two is a common early pitfall on Cisco UCS hardware.

Phase 4 — MicroCloud Software Installation

Objective: install the MicroCloud stack and its components on all three servers.

Key Steps

1. Install the four snap packages on every server:

```
sudo snap install microcloud --channel=candidate
sudo snap install microceph --channel=candidate
sudo snap install microovn --channel=candidate
sudo snap install lxd --channel=latest/stable
```

2. Verify the installed versions on each node:

```
snap list
```

NOTE At deployment time, MicroCloud, MicroCeph, and MicroOVN were installed from the candidate channel as the stable channel was not yet available for these components. If an installation times out due to connectivity, simply re-run the command — the download resumes.

Phase 5 — Cluster Initialization

Objective: form the cluster, configure distributed storage and networking, and join all three nodes — the core of the deployment.

Key Steps

1. On the two secondary nodes, start the join process so they become discoverable:

```
sudo microcloud join
```

2. On the initiating node (SVR01), start the guided initialization:

```
sudo microcloud init
```

3. Follow the interactive wizard: select the cluster network interface, detect and add the other nodes, enable distributed storage (Ceph) and select the available disks, enable distributed networking (OVN) and define the instance address range and gateway.

4. Verify the cluster formed correctly:

```
microcloud status
lxc cluster list
sudo microceph status
```

 **INSERT SCREENSHOT HERE**

Capture: output of 'lxc cluster list' showing SVR01, SVR02, SVR03 with their roles and ONLINE status.

Figure 5. Cluster members — all three nodes joined and online.

KEY LESSON For cluster-wide operations, run commands sequentially rather than chaining them. In this version, secondary nodes require an explicit join command rather than passive discovery alone.

Phase 6 — Management Interface & Verification

Objective: enable the LXD web interface for graphical management and verify overall cluster health.

Key Steps

1. Enable the LXD HTTPS listener on the cluster:

```
sudo lxc config set core.https_address :8443
```

2. Generate a trust token and use it to authenticate the browser via a client certificate:

```
lxc auth identity create tls/lxd-ui --group admins
```

3. Access the interface at <https://10.0.0.11:8443> and complete certificate-based login.
4. Run the full suite of health checks (cluster, storage, networking).

□ INSERT SCREENSHOT HERE

Capture: the LXD UI dashboard after login, showing the cluster/instances view.

Figure 6. LXD web interface — graphical cluster management.

Phase 7 — Monitoring & Observability

Objective: deploy a complete monitoring stack so every node's health and capacity is visible in real time.

Key Steps

1. Deploy Prometheus and Grafana as containers on the cluster, each with a dedicated network interface for connectivity.
2. Install Node Exporter on all three physical hosts to expose system metrics:
`node_exporter & # later converted to a systemd service`
3. Configure Prometheus to scrape all three nodes on port 9100.
4. Connect Grafana to Prometheus and import monitoring dashboards, including a custom multi-node operations dashboard.
5. Expose Prometheus and Grafana externally via LXD proxy devices.

□ INSERT SCREENSHOT HERE

Capture: the full Mission Control Grafana dashboard with live data (fleet stats, heartbeat, per-node vitals).

Figure 7. AFRITEC Mission Control — real-time cluster monitoring dashboard.

KEY LESSON Background processes started manually do not survive a reboot. All monitoring services were subsequently converted to managed system services, and the monitoring containers set to auto-start, so the entire stack recovers automatically after any restart.

5. Operations & Maintenance

5.1 Daily Health Checks

The following commands, run on the leader node, provide a complete health snapshot of the platform:

```
microcloud status      # Overall cluster health
lxc cluster list       # Node membership and status
sudo ceph -s           # Storage health and capacity
lxc list               # Running instances
```

5.2 Graceful Shutdown Procedure

Order: stop workload containers first, then power down the voter nodes, then the leader last.

Order	Action	Command
1	Stop containers (on leader)	sudo lxc stop grafana; sudo lxc stop prometheus
2	Power off SVR03	ssh ...svr03 → sudo poweroff
3	Power off SVR02	ssh ...svr02 → sudo poweroff
4	Power off SVR01 (last)	sudo poweroff

5.3 Graceful Startup Procedure

Order: power on the leader first to establish the cluster database, then the voters to form quorum. Monitoring services recover automatically.

Order	Action	Notes
1	Power on SVR01 (leader)	Wait 2–3 min to fully boot
2	Power on SVR02	Wait for boot
3	Power on SVR03	Wait for boot
4	Verify HEALTH_OK	Allow 3–5 min for quorum

QUORUM REMINDER Ceph requires at least two of three nodes online before its database fully starts. A single node alone cannot form a healthy cluster — this is expected behavior, not a fault.

5.4 Secure Remote Access

The platform is managed remotely through Tailscale, an encrypted mesh VPN. The leader node advertises the cluster subnet, allowing authorized devices to securely reach all management interfaces (LXD UI, Grafana, Prometheus) from outside the office network, without exposing any service to the public internet.

□ INSERT SCREENSHOT HERE

Capture: the Tailscale admin console showing connected devices (laptop + node), or the dashboard loaded remotely.

Figure 8. Secure remote access — Tailscale mesh connecting laptop and cluster.

BEST PRACTICE For resilience, install the remote-access agent on more than one node. If only the leader provides the remote path and it is offline, remote access is lost until it is restored. Multiple entry points eliminate this single point of failure.

6. Results & Current State

The deployment achieved a fully operational, monitored, high-availability private cloud. The cluster reports healthy status across all core subsystems.

Subsystem	Status	Detail
Cluster	OPERATIONAL	3 nodes online, quorum established
Storage (Ceph)	HEALTH_OK	17 OSDs, ~19 TiB available
Networking (OVN)	OPERATIONAL	Software-defined tenant networking active
Compute (LXD)	OPERATIONAL	Ready for VMs & containers
Monitoring	OPERATIONAL	All nodes reporting; dashboards live
Remote Access	OPERATIONAL	Secure mesh VPN configured

6.1 Outstanding Item

SVR01 storage expansion: the leader node's additional disks are currently presented as RAID virtual drives rather than individual JBOD devices, and so are not yet contributing to the Ceph pool. The cluster is fully operational at 17 OSDs without them. Adding this capacity is a planned enhancement that will increase total storage without affecting availability.

6.2 Planned Enhancements

- Add SVR01 disks to the Ceph pool to expand storage capacity.
- Implement security hardening: SSH key-only authentication, two-factor authentication for privileged commands, and least-privilege access controls.
- Deploy the remote-access agent on all nodes for redundant secure access.
- Establish a formal backup strategy for configurations and critical data.

Annex A — Troubleshooting Guide

This annex documents the real issues encountered during deployment and operation, with their root causes and resolutions. Each entry is written so that a future engineer facing the same symptom can diagnose and resolve it quickly.

A.1 Storage Controller Not Visible to the Operating System

Symptom. During OS installation on one server, the installer reported no disks found, even though CIMC showed all disks healthy.

Root Cause. The RAID controller card was communicating with CIMC over the management bus, but its data connection to the CPU was not initializing — making it invisible to the operating system. Software-level fixes (BIOS defaults, firmware settings) did not resolve it.

Resolution. Physically reseal the RAID controller card. Power down, remove and firmly re-insert the card, then power on. The disks became visible immediately and all prior configuration was retained.

A.2 Operating-System Disk Not First in Device Order

Symptom. The OS virtual drive appeared as the last device instead of the first, risking installation to the wrong disk.

Root Cause. When RAID virtual drives and JBOD disks coexist on one controller, the individual JBOD disks are enumerated first; the virtual drive appears afterward.

Resolution. Identify the correct install target by its size and model, not by its device letter. Select only the OS virtual drive; never use 'erase all disks.'

A.3 Bootloader Failure After Installation

Symptom. A server failed to boot from its OS drive after installation.

Root Cause. The bootloader was not correctly installed to the system partition.

Resolution. Boot from the installation USB in live mode, install and run a boot-repair utility, and apply the recommended repair. The system then booted normally.

A.4 Servers Have No Network After Installation

Symptom. None of the servers could reach the network or each other after OS installation.

Root Cause. The IP addresses in use belonged to the CIMC management controllers, which are on a separate network from the operating system. The OS data ports had no cables connected.

Resolution. Connect physical cables to the data-plane ports, then assign static IP addresses via Netplan on the correct interfaces. Verify all nodes can ping each other.

A.5 Software Package Installation Fails

Symptom. Installing the cloud packages failed with a network-unreachable error, although local updates had worked.

Root Cause. The package repository requires direct internet access; the gateway or DNS was not correctly configured, or the upstream link was briefly unavailable.

Resolution. Verify internet connectivity, correct the gateway and DNS settings in Netplan if needed, and re-run the installation — downloads resume automatically.

A.6 Cloud Package Not Available on Stable Channel

Symptom. The MicroCloud package could not be installed from the default stable channel.

Root Cause. At the time of deployment, MicroCloud was not yet published to the stable channel.

Resolution. Install MicroCloud, MicroCeph, and MicroOVN from the candidate channel. LXD remains on its stable channel.

A.7 Leader Node Disks Not Available to Storage

Symptom. The leader node's additional disks were not offered as available for the storage pool.

Root Cause. The controller presented these disks as combined RAID volumes rather than individual pass-through devices.

Resolution. Reconfigure the disks in CIMC as individual JBOD devices, then add them to the storage pool. (Tracked as an outstanding enhancement; the cluster operates fully without them.)

A.8 Container Network Creation Fails in Cluster Mode

Symptom. Creating a bridge network for containers failed in the clustered environment.

Root Cause. In a cluster, networks must first be declared on each member node individually, then finalized cluster-wide. Chaining these commands consumed the pending state too quickly.

Resolution. Run the per-node declaration commands sequentially, one at a time, then issue the final creation command with the network properties.

A.9 Container Networking Resets on Restart

Symptom. Monitoring containers lost network connectivity every time they restarted.

Root Cause. Runtime network settings applied by command live only in memory and are cleared on restart.

Resolution. Make the configuration persistent by writing a Netplan file inside each container defining its address, route, and DNS. Settings then survive every restart.

A.10 Monitoring Configuration Errors from Manual Entry

Symptom. The metrics service failed to start due to configuration formatting errors.

Root Cause. Configuration files typed manually into a remote console are prone to indentation errors, which the service cannot parse.

Resolution. Write the entire configuration file in one operation using a here-document rather than editing line by line, eliminating indentation mistakes.

A.11 Service Binary Name Changed Between Versions

Symptom. The dashboard service would not start using its documented binary name.

Root Cause. A newer release renamed the executable and restructured its startup command.

Resolution. Use the current binary name and subcommand form. Verify actual paths and names on disk rather than relying on older documentation.

A.12 Management Interface Certificate Transfer

Symptom. The web-interface trust token could not be pasted into the browser from the remote console (no clipboard support).

Root Cause. The remote console does not synchronize the clipboard with the workstation.

Resolution. Write the token to a file, serve it briefly over a simple local HTTP server, retrieve it from the workstation browser, then stop the server. This bridge technique is reusable for any text transfer where copy-paste is unavailable.

A.13 Containers Cannot Reach the Internet

Symptom. Service containers could not reach external addresses.

Root Cause. The primary virtual network does not provide outbound internet routing for containers by default.

Resolution. Create a dedicated bridge network with address translation and attach it to the containers as a second interface, providing outbound access.

A.14 Dashboard Import Fails Without Internet

Symptom. Importing a dashboard by ID failed because the service could not reach the external dashboard repository.

Root Cause. The import feature requires the server process itself to reach the internet, which the container could not do reliably.

Resolution. Download the dashboard definition on a connected host, serve it locally, and import it by pasting the definition directly.

A.15 Monitoring Shows No Data After Restart

Symptom. After a reboot, the dashboards displayed no data.

Root Cause. The monitoring processes had been started manually and did not restart with the system; container networking had also reset.

Resolution. Convert all monitoring processes to managed system services, make container networking persistent, and enable container auto-start. The full stack now recovers automatically after any restart.

A.16 Remote Access Lost After Reboot

Symptom. The cluster became unreachable from outside the office following a reboot, although the leader node was powered on.

Root Cause. IP forwarding on the leader node was disabled after the reboot, so it could no longer route remote traffic to the other nodes. The remote-access route also required re-advertising.

Resolution. Connect to the leader over the direct mesh address, re-advertise the subnet route, and make IP forwarding persistent so it survives future reboots. Disable system sleep to prevent the node dropping offline.

Annex B — Glossary

Term	Definition
IaaS	Infrastructure as a Service — on-demand virtualized compute, storage, and networking.
MicroCloud	Canonical's integrated private-cloud stack bundling LXD, Ceph, and OVN.
LXD	A manager for Linux containers and virtual machines.
Ceph / MicroCeph	A distributed storage system pooling disks across servers into one resilient cluster.
OSD	Object Storage Daemon — a Ceph process managing a single physical disk.
OVN / MicroOVN	Open Virtual Network — software-defined networking for virtual workloads.
CIMC	Cisco Integrated Management Controller — out-of-band server management interface.
RAID	A method of combining disks for performance or redundancy.
JBOD	'Just a Bunch of Disks' — pass-through mode presenting disks individually to the OS.
Quorum	The minimum number of cluster nodes that must agree for the cluster to operate.
Prometheus	A metrics collection and time-series database system.
Grafana	A visualization platform for building monitoring dashboards.
Node Exporter	An agent exposing a server's system metrics to Prometheus.
Netplan	Ubuntu's network configuration system using YAML files.
Snap	A self-contained software package format used by Canonical.
Tailscale	An encrypted mesh VPN based on the WireGuard protocol.

END OF DOCUMENT This report documents the complete lifecycle of the AFRITEC private cloud platform. It is intended as both a record of the engagement and a reproducible guide for deploying an equivalent platform on comparable infrastructure.